Sparse Machine Learning Discovery of Dynamic Differential Equation of an Esophageal Swallowing Robot

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***Abstract*—Considering the limitation of conventional rheometry measurements and the relevance of the study of food viscosity in the study of dysphagia, we commenced developing a soft-bodied esophageal swallowing robot (ESR). Using experimental data, this paper aims to discover the differential equations (DEs) of the ESR that governs the peristaltic deformation in the esophageal conduit at the given time-varying pressure. The deformation data of the conduit are collected from a quarter version of the ESR due to the inaccessibility to the esophageal occlusion. The three-dimensional displacements of the markers placed on the interior surface of the conduit are measured using a Vicon optical motion capturing system. The dimension of the dataset is first reduced using the principal orthogonal decomposition (POD) method, which is then used to discover the ESR's DEs by the sparse identification of nonlinear dynamics (SINDy). The marker displacements are considered as the ESR's states. The ESR's states are reduced from 27 to 2 by initially applying power spectral density (PSD), and then the POD. The identified states essential to the DEs capture 95% of the total variance of the deformation dataset. Finally, the conduit deformation simulated from the DEs are validated by the experimental measurements in the full version of the ESR.**

***Index Terms*—Discovery of differential equations (DEs), machine Learning (ML), soft robotics, swallowing robot.**

I. Introduction

MACHINE Learning (ML) is fast becoming a key instrument for understanding the underlying physics of the nonlinear dynamical systems as researchers are collecting data from experiments. In general, ML approaches are highly capable to map arbitrary functions. Two crucial contributions to the revelation of physical laws from the collected experimental data can be found in [1] and [2]. The computational method discussed in [1] used motion capture data from various systems to demonstrate the discovery of the physical laws that are governing those systems.

In 2016, Brunton *et al.* [3] described a sparse regression based technique known as the sparse identification of nonlinear dynamics (SINDy). It has become dominant in identifying the dynamics of linear and nonlinear oscillators, simple canonical systems, the Lorenz system, and the fluid vortex shredding behind an obstacle. Despite the efficacy of the SINDy in [3], it is ill-suited for higher order dynamical systems [4].

A low-dimensional representation of such systems can often be leveraged by the application of reduced-order model (ROM) based techniques such as the total principle component regression (TPCR) [5], and the principle orthogonal decomposition (POD) based on the singular value decomposition (SVD) [3]. POD facilitates reduced-order models of the physical system such that the maximum variance of the data collected from the system is captured in the model. The principal component regression (PCR) models the relationship between a dependent variable and one or more independent variables. Instead of directly implementing a standard linear regression model, the PCR uses principle component analysis for estimating unknown regressing model coefficients. Compared to the PCR, the TPCR decomposes the output space into two subspaces. In [5], a TPCR-based model was developed for data-driven key performance indicators oriented fault detection. For reliable online tracking ability, and availability of new measurements online, the TPCR model needs to be updated (online training) by solving SVD repeatedly, which increases the computational burden significantly. The recursive-TPCR implements recursive manner update by performing large-scale matrix calculations in the singular value (SV) space instead of the measurement space [5]. In this paper, the model concerned is an offline model, so it has not been updated online, and, hence, the SVD has been applied only once to map the measurement data to a subspace, where it can be best represented.

Soft robotics, which emerges from the notion of natural and biological phenomena where robots are designed and developed by employing soft and compliant modules [6], is an increasingly important area in the field of robotics. A soft-bodied robot is either a single soft body or a set of soft bodies, which has a large number of DOFs and involves strong nonlinear material behaviors and large deformation. Its movement or shape change or reconfiguration is achieved via distributed pneumatic net or

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electroactive polymer actuators. The dynamics of such a robot can hardly be derived from the first principles in state space. Soft robots manufactured from soft elastomers like silicone rubber are not very well understood due to their continuous, complex, and highly compliant intrinsic deformation. As soft robotics is emerging, the theoretical foundation is yet to be laid for kinematics, dynamics, and control of soft robots [7], [8]. The dynamic modeling of soft robots can open the gateway for new capabilities of robots. The models can be used to develop control and planning for soft robotic systems [9]. The modeling of a soft robot plays an important role in developing compliance, when the robot interacts with the environment.

In recent years, there has been increasing interest in the study of human esophageal swallowing process and its disorders [10]. Dysphagia, a swallowing disorder, affects a significant proportion of the elderly population [11]. One of the management methods is to provide foods with modified viscosities. The process of peristalsis transports the food bolus through the esophagus to the stomach. However, the complex relationship between the bolus formulation and its effect on the peristalsis is tough to grasp. This difficulty is aided by the challenge of performing *in vivo* bolus swallow experiments due to ethical concerns [12].

The past decade has seen a rapid development in swallowing robots, typically with application in biomimicry and medical training. Kobayashi *et al.* [13] have focused on the oral and pharyngeal stages of swallowing. Noh *et al.* [14] developed a biologically inspired device that has successfully modeled the swallowing process concerning medical imaging analysis. It is actuated by servo motors, where wire assemblies were used to transmit the displacement actuation. Miki *et al.* [15] designed a prototype of an artificial esophagus with a conduit diameter of 18 mm. The prototype is built up of discrete units. Each unit is actuated by applying current through the shape-memory alloy spring, resulting it to heat up and shrink. Together, these devices provide an insight in the field of swallowing, but they have different limitations.

Soft robotic techniques provide compliance and novel shapes to robots. However, the methods have received little attention in the field of swallowing robots. The combination of pneumatic actuation and soft elastomers promotes design without skeletal structures, which are necessary for mimicking the swallowing process. The limitation of the conventional rheometry measurements [12], and the relevance of the study of food viscosity has motivated the development of an esophageal swallowing robot (ESR), inspired by the human swallowing phenomena. The ESR was developed as a medical instrument to simulate *in vitro* the bolus transition in esophagus in a biologically faithful manner. By means of the ESR, the rheological properties and transport efficacy of the food bolus can be evaluated for the purpose of dysphagia management. During swallowing, by measuring the physical interaction of the food bolus with the ESR's conduit wall could provide data about the bolus travel experience from the perspective of human recognition.

FEA-based modeling methods are predominant in soft robotics, and primarily used for analysis of the system's dynamic behaviors. They can be used to generate nonlinear materials model of hyperelasticity. These FEA methods are not

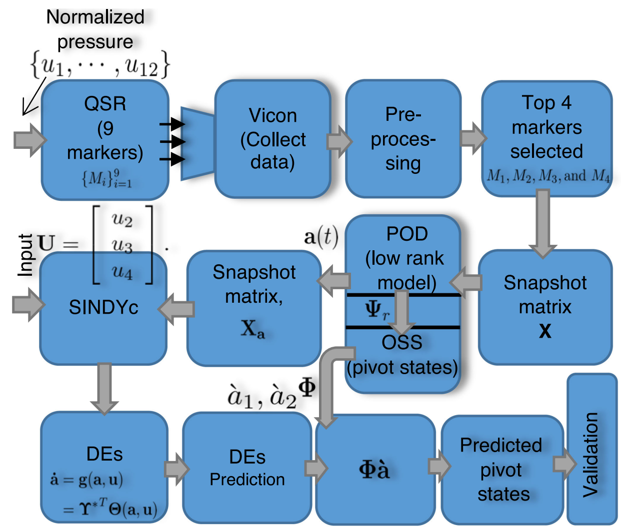


Fig. 1. Block diagram summarizing the procedure of ESR differential equations (DEs) discovery.

applicable for the real-time dynamics control of either a rigid or a soft robot. The previous work conducted by Dirven *et al.* [12] and Zhu *et al.* [16] was positive, but they could not explicitly define the underlying physics governing the ESR. In [17], Chen *et al.* proposed a nondynamical ESR model, which was able to predict the conduit deformation for a given input pressure. The discussed model can only be used for structural and performance improvement and not for control. To understand the kinematic and dynamic features of the ESR, and for closed-loop control, data-driven ML methods can provide some great potential for modeling the ESR.

The main aim of this paper is to discover the governing DEs of the ESR from the data collected from the experiments. The DEs capture the ESR's peristaltic conduit deformation with respect to the applied pressure at any instant of time. The input pressure signals can be actual pressure values with a physical unit or abstract numerical values. In the process of the DE discovery, four significant steps were followed. First, due to the inaccessible conduit deformation of the ESR, a quarter version of the swallowing robot (QSR) was developed in order to collect the data. The QSR is used to observe and model the occlusive motion of the conduit of the ESR. Second, by using the Vicon system, the QSR's conduit deformation data were collected by tracking three-dimensional marker movements, placed on the conduit of the QSR. Third, the size of the collected dataset was reduced using the POD, and the SINDy was applied on the reduced dataset to determine the governing DEs. Finally, the conduit deformation calculated from the DEs discovered was validated by the experimental measurements in the full version of the ESR. Fig. 1 provides the block diagram that summarizes the procedure of the DEs discovery. The proposed method can be seen as a combination of structure and parameter identification because the set of functions and their corresponding coefficients in the ESR's DEs are determined by the data-driven modeling.

This paper makes several noteworthy contributions to the area of soft robotics. To the best of our knowledge, this is the

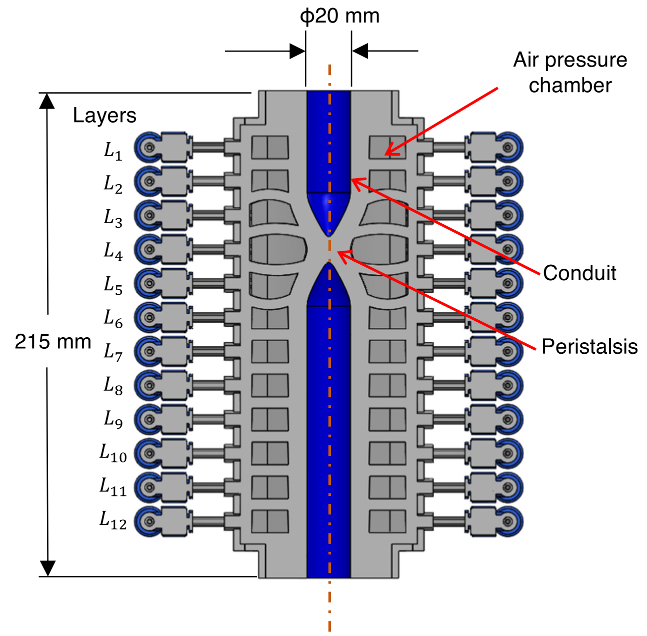


Fig. 2. Schematic of the ESR.

first time that ML has been applied to identify a structural and parametric dynamical model of the ESR. The methods reveals the underlying dynamics of the soft-bodied ESR in the form of DEs model. In contrast to the earlier models of the ESR, the mathematical model (DEs model) presented in this literature is dynamical. Since ML solely depends on the captured data, the discussed procedure is not robot specific and does not depend upon the actuation system and morphology of the robot. The procedure in this paper is general, while the ESR is a case study. It can be extended to any soft-robotic system, provided input–output dataset can be generated. The DEs discovered in this paper also explicitly define the nonlinear behavior of the robot. The analysis of the DEs has extended our knowledge about the deformation of the ESR.

II. Problem Statement

The soft-bodied ESR, as shown in Fig. 2, was constructed from vulcanizing a silicone rubber at room temperature [12]. The conduit of the robot has diameter of 20 mm and length of 215 mm, which can be actuated pneumatically. It consists of 12 layers (L_1, L_2, …, L_{12}) of four equally spaced air chambers (whorl of chambers) at each level. The actuation protocol incorporates the inflating of all the chambers in a whorl with the same pressure. The conduit and its surrounding actuation region is devoid of any rigid element, which is analogous to the biology of the esophagus. Corresponding to its biological counterpart, the robot is able to provide a continuously deformable peristaltic occlusion in its conduit, manifesting distributed actuation and compliance by inflating four chambers layer by layer with the pressure, regulated via a proportional valve.

To witness the deformation of the chambers, a stretchable sensor array was put inside the conduit, and experiments were conducted on it. From the experiments, a nonlinear phenomenon

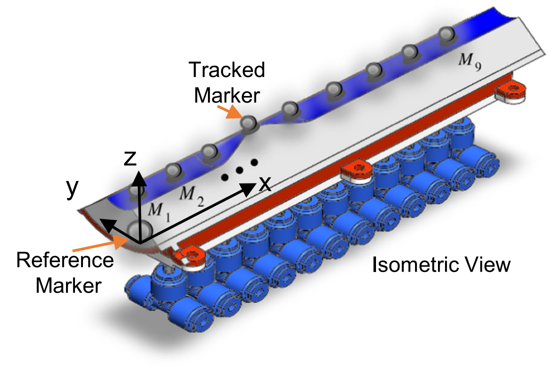


Fig. 3. Schematic of the QSR.

between the chamber deformation and the applied time-varying pressure trajectory was reported [16]. The nonlinearity occurs due to material continuous intrinsic deformation, and dependencies between the input trajectories. There does not exist any first principle based method that is able to describe the nonlinear relationship between the applied pressure and the conduit deformation. Hence, in this paper, several ML methods were applied to discover the DEs of the ESR, which solely relies on the experimental data. The DEs define the ESR conduit's peristaltic deformation with the time-dependent applied pressure profile.

The ESR has limited visibility inside its conduit, which challenges the collection of the experimental data needed for the DE discovery. To address this issue, a quarter version of the ESR was developed, and its conduit deformation was measured using a Vicon optical motion capture system. It consists of infrared cameras that track retroreflective markers located on an object. Fig. 3 illustrates the quarter swallowing robot (QSR) in which the location of the markers can be seen. One marker was placed at the center of the air chamber of each layer so that the chamber deformation can be tracked. Hence, in total, nine markers were placed nine 9 layers of the QSR, and their time-series movement in the x-, y-, and z-axes were measured.

A dataset of dimension $m = 5408$ and $n = 27$ ($n = 27$ corresponds to the x-, y-, and z-axes movement of the nine markers located in Fig. 3 was collected from the experiments, where m represents the number of snapshots (time samples) and n the number of marker states. The number of snapshots (m) was selected in such a way that a deformation dataset of five peristalsis cycle could be generated. Of the five cycles, three cycles were used for training and two for validating the DEs. Hence, a ratio of 60:40 of m was used for training and validation. Each snapshot was taken at a sampling interval (T_s) of 0.01 s.

In order to determine the DEs, the x-, y-, and z-axes marker deformation were defined as the system states, and it was assumed that the states were observable because all the markers are trackable. Since the dimension of the states were 27, and since the SINDy does not converge for high state dimensional system, so it was not possible to determine the underlying DEs from the captured dataset in the current space. By determining and observing the power spectral density (PSD) of the nine marker states, the first four markers' states (12 states) were selected. In this paper, the POD [3] and Galerkin projection [18] were applied to project the markers M_1, M_2, M_3, and M_4 dataset to a new subspace (S). The subspace has been defined by r vector modes embedded in the columns of $\mathbf{\Psi}_r$. These r vector modes are the basis of S. Since, the POD gives a principle way to do a low-rank truncation of the data, so $r << 12$.

Once the reduced number of states, $\mathbf{a}(t)$, was obtained, the SINDy was applied to find the DEs of the ESR. It is a widely held view that most dynamical complex systems are governed by a few nonlinear terms, and, thus, sparsity occurs in the high-dimensional nonlinear function space [3]. The DEs to be discovered are in the form of

$$\frac{d\mathbf{a}(t)}{dt} = \mathbf{g}(\mathbf{a}(t), \mathbf{u}(t)) \tag{1}$$

where $\mathbf{u}(t)$ is the applied pressure trajectory and $\mathbf{g}(.)$ a nonlinear function consisting of constant and polynomial terms of $\mathbf{a}(t)$ and $\mathbf{u}(t)$.

III. Reduced-Order Modeling

A. Reduction of High-Dimensional DEs

Let $\mathbf{x}(t)$ be the state of the ESR that lies on an n-dimensional hyperspace, $\mathbb{R}^n$, such that $n \gg 1$. In the case of the ESR, $n = 27$ represents the number of marker states. Let S be a set of r linearly independent orthonormal vector modes such that $S = \{\boldsymbol{\psi}_1, \boldsymbol{\psi}_2, \ldots, \boldsymbol{\psi}_r | \boldsymbol{\psi}_j \in \mathbb{R}^n \& ||\boldsymbol{\psi}_j|| = 1\}$. The linear combination of these vector modes spans the r-dimensional subspace S in $\mathbb{R}^n$, and, hence, S can also be represented by the column space of $\mathbf{\Psi}_r$ ($C(\mathbf{\Psi}_r)$). The vector modes stored in $\mathbf{\Psi}_r$ are the basis of S. For the ESR, S represents the subspace where the projection of the marker dataset lies. The dimension of S given by r is much smaller as compared to the dimension of the hyperspace. Let $\mathbf{x}(t) \notin S$, and, hence, no linear combination of these modes can represent $\mathbf{x}(t)$. Let $\tilde{\mathbf{x}}(t)$ be the orthogonal projection of $\mathbf{x}(t)$ on S then, $\tilde{\mathbf{x}}(t)$ can be written in terms of linear combination of the following modes [19]:

$$\mathbf{x}(t) \approx \tilde{\mathbf{x}}(t) = \sum_{j=1}^{r} a_j(t)\boldsymbol{\psi}_j = \mathbf{\Psi}_r \mathbf{a}(t) \tag{2}$$

where $\mathbf{\Psi}_r = [\boldsymbol{\psi}_1, \boldsymbol{\psi}_2, \ldots, \boldsymbol{\psi}_r]$ and $\mathbf{a}(t) = [a_1(t), a_2(t), \ldots, a_r(t)]^T$. If $r \ll n$, then, by knowing r values of $a_1(t), a_2(t), \ldots, a_r(t)$, a close estimate of $\mathbf{x}(t)$ represented by $\tilde{\mathbf{x}}(t)$ can be determined. The coefficient $a_j(t)$ associated with each ψ_j represents the new reduced states of the ESR in S. By applying vector projection theory on $\mathbf{x}(t)$, it can be written as

$$\mathbf{x}(t) \approx \tilde{\mathbf{x}}(t) = \mathbf{\Psi}_r \mathbf{\Psi}_r^T \mathbf{x}(t) \tag{3}$$

where $\mathbf{\Psi}_r^{\dagger} = \mathbf{\Psi}_r^T$ because $\mathbf{\Psi}_r$ is an orthonormal matrix.

Now, consider a nonlinear dynamical system with input $\mathbf{u}(t)$ (such as the ESR) is governed by a set of the following first-order DEs:

$$\frac{d\mathbf{x}}{dt} = \mathbf{f}(\mathbf{x}(t), \mathbf{u}(t)) \tag{4}$$

where $\mathbf{u}(t)$ is the control input lying in $\mathbb{R}^l$ and $\mathbf{f}$ any nonlinear continuously differentiable function that maps: $\mathbf{f} : \mathbb{R}^n \times \mathbb{R}^l \rightarrow \mathbb{R}^n$. With $n = 27$ being the number of the ESR's states, it is very difficult to obtain $\mathbf{f}$ directly from $\mathbf{x}(t)$. Hence, substituting (2) and (3) in (4), we can rewrite (4) as

$$\frac{d\mathbf{a}}{dt} = \mathbf{\Psi}_r^T \mathbf{f}(\mathbf{\Psi}_r \mathbf{a}(t), \mathbf{u}(t)), \mathbf{a}(0) = \mathbf{\Psi}_r^T \mathbf{x}(0) \tag{5}$$

which is the ROM of the original dynamics of the ESR in (4). It consists of r DEs, which describe the dynamics of the time evolution of $\mathbf{a}(t)$ and lie in S.

The result of (5) has shown a successful projection of the dynamics of the n-dimensional ESR to a r-dimensional subspace, where it is possible to discover the DEs. By comparing (5) and (1), $\mathbf{\Psi}_r^T \mathbf{f}(.)$ can now be represented as $\mathbf{g}(.)$, and the DEs can be discovered by determining $\mathbf{g}(.)$. Once the DEs are determined, $\tilde{\mathbf{x}}(t)$ can again be reconstructed from the predicted data generated from the simulated DEs by applying $\tilde{\mathbf{x}}(t) = \mathbf{\Psi}_r \mathbf{a}(t)$. In this way, tremendous amount of computational efficiency can be achieved [20]. Hence, r number of DEs can be determined by applying Galerkin projection to a system of n nonlinear DEs such that $r << n$.

B. Data-Dimension Reduction

The dimension of S and, hence, r can be determined by applying the POD on the collected marker dataset (ESR's states) [18]. To apply the POD, first the snapshots of all the ESR's states are collected in $\mathbf{X} = [\mathbf{x}(t_1), \mathbf{x}(t_2), \ldots \mathbf{x}(t_m)] \in \mathbb{R}^{n \times m}$. The vector $\mathbf{x}(t_j)$ is the state of the ESR at time $t = t_j$ and is given by $\mathbf{x}(t_j) = [x_1(t_j), x_2(t_j), \ldots, x_n(t_j)]^T$.

The snapshot matrix $\mathbf{X}$ belongs to a high-dimensional subspace of $\mathbb{R}^n$, and as $n \ll m$, truncated SVD can be applied to decompose $\mathbf{X}$ into three matrices $\mathbf{\Psi}_r$, $\mathbf{\Sigma}_r$, and $\mathbf{V}_r$, written as follows [18]:

$$\mathbf{X} = \begin{bmatrix} | & | & \cdots & | \\ \mathbf{x}_1 & \mathbf{x}_2 & \ldots & \mathbf{x}_m \\ | & | & \cdots & | \end{bmatrix} \approx \tilde{\mathbf{X}} = \mathbf{\Psi}_r \mathbf{\Sigma}_r \mathbf{V}_r^T. \tag{6}$$

The $\mathbf{\Psi}_r = [\boldsymbol{\psi}_1, \ldots, \boldsymbol{\psi}_r]$ from the truncated SVD is an $n \times r$ matrix that contains the left singular vectors, which are also known as the POD modes. These r POD modes are an orthonormal set of vectors in which most of the dynamics of the ESR were embedded. The truncated SVD gives a principal way to do a low-rank truncation of $\mathbf{X}$ where a low-rank or low-dimensional embedding can be used to represent $\mathbf{X}$. The SVs $\{\sigma_1, \cdots, \sigma_r\}$ in the diagonal of $n \times r$ diagonal matrix $\mathbf{\Sigma}_r$ describe the importance of each orthonormal vector (direction). The right singular vectors embedded in the $r \times m$ matrix $\mathbf{V_r}^T$ describe the corresponding time dynamics of each mode such that $\mathbf{a}(t) = \mathbf{\Sigma}_r \mathbf{V_r}^T$.

C. States Essential to DEs

Let $p \leq n$ be the number of states required in $\mathbf{x}$ to define the dynamics of the ESR, and let the rest of the $n - p$ states can be obtained by the linear combination of the p states. Let $\mathbf{y} \in \mathbb{R}^p$ be a tracking vector that contains the p states. Let $\mathbf{C} \in \mathbb{R}^{p \times n}$ is a selection matrix such that $\mathbf{y} = \mathbf{C}\tilde{\mathbf{x}}$. The value of $\mathbf{C}$ provides the location of the p states that are needed to be tracked. By using (2) in $\mathbf{y} = \mathbf{C}\tilde{\mathbf{x}}$, it can be written as

$$\mathbf{y} = \mathbf{C}\mathbf{\Psi}_r\mathbf{a} = \mathbf{\Phi}\mathbf{a} \tag{7}$$

which, in general, is overdetermined if $p > r$, but if $p = r$, then the solution is unique. Let $\varrho = \{\varrho_1, \varrho_2, \ldots, \varrho_p\} \subseteq \{1, 2, \ldots, n\}$ denote the p pivot states indices. Depending upon the system, $\varrho_j \in \varrho$ can take any value from $j = 1, \ldots, n$; hence, the optimal state vector is $\mathbf{x} = [x_{\varrho_1}, x_{\varrho_2}, \ldots, x_{\varrho_p}]^T$. The structure of matrix $\mathbf{C}$ is governed by the set ϱ where all the elements in every row of $\mathbf{C}$ are zero except the ϱ_j^{th} entry, which is equal to one.

If $\hat{\mathbf{a}}$ is the best estimate of $\mathbf{a}$ in terms of least square sense in (7), then $\hat{\mathbf{a}}$ can be written as

$$\hat{\mathbf{a}} = \begin{cases} \mathbf{\Phi}^{-1}\mathbf{y} = (\mathbf{C}\mathbf{\Psi}_r)^{-1}\mathbf{y}; p = r \\ \mathbf{\Phi}^{\dagger}\mathbf{y} = (\mathbf{C}\mathbf{\Psi}_r)^{-\dagger}\mathbf{y}; p > r \end{cases}. \tag{8}$$

Let $\hat{\mathbf{x}}$ be an approximated version of $\tilde{\mathbf{x}}$, which can be generated by applying $\hat{\mathbf{x}} = \mathbf{\Psi_r}\hat{\mathbf{a}}$. By applying optimal state selection (OSS), the l_2 error (least square error) between $\mathbf{a}$ and $\hat{\mathbf{a}}$ can be minimized. The OSS problem deals with obtaining $\mathbf{C}$ such that the optimum rows of $\mathbf{\Psi_r}$ can be determined. Hence, the OSS algorithm maps the r-dimensional reduced subspace S in $\mathbb{R}^n$ to an r-dimensional subspace $S_p \in \mathbb{R}^r$ in $\mathbb{R}^p$ [19], [20].

Matrix $\mathbf{C}$ can be determined by finding out the first p pivot states from n states in such a way that $\mathbf{a}$ can be recovered from $\mathbf{y}$. By applying QSR decomposition with column pivoting on $\mathbf{\Psi}_r\mathbf{\Psi}_r{}^T$ ($\mathbf{\Psi_r^T}$) for $p > r$ ($p = r$), the pivot states indices ϱ can be determined, and $\mathbf{C}$ can be structured in accordance with it [19], [20].

IV. Discovery of DEs for the Robot

In Section III, by applying ROM-based techniques, the states $\mathbf{x}(t)$ of the ESR have been reduced to $\mathbf{a}(t)$ in order to derive the DEs of the ESR given by (1). This section generalizes the SINDy with control (SINDYc) method to incorporate the time-varying air-pressure input to the pneumatic chambers of the ESR that are actuated sequentially [21], [22]. The SINDYc was applied to the snapshots of $\mathbf{a}(t)$ and $\mathbf{u}(t)$ by generating a data library matrix, which is discussed further below in this section.

It was observed that the DEs modeling of the dynamics of any physical system comprises a combination of few linear and nonlinear terms [3]. Hence, these terms are called active terms. Based on such assumption, the SINDYc method was implemented, which balances the number of the active terms in the DEs with accuracy and overfitting, resulting in a parsimonious model of the ESR.

An input snapshot matrix $\mathbf{U} \in \mathbb{R}^{l\times m}$, similar to $\mathbf{X}$, is prepared, where l represents the number of the inputs to the ESR. Since it is difficult to make the SINDYc algorithm converge with the actual ESR's states, another snapshot matrix $\mathbf{X}_a$ is introduced. The snapshot matrix $\mathbf{X}_a$ having a dimension of $r \times m$ is structurally similar to $\mathbf{X}$, and it contains the snapshots of $\mathbf{a}(t)$ instead of the snapshots of $\mathbf{x}(t)$.

In order to find the active terms that can define the governing DEs of the ESR, the following library matrix $\mathbf{\Theta}(\mathbf{X}_a, \mathbf{U})$ was designed:

$$\mathbf{\Theta}(\mathbf{X}_a, \mathbf{U}) = \begin{bmatrix} 1 \; \mathbf{X}_a \; \mathbf{U} \; \mathbf{X}_a \otimes \mathbf{X}_a \; \mathbf{U} \otimes \mathbf{U} \cdots \end{bmatrix}. \tag{9}$$

Matrix $\mathbf{\Theta}(\mathbf{X}_a, \mathbf{U})$ consists of the candidates of nonlinear function as well as nonlinear cross terms of $\mathbf{a}$ and $\mathbf{u}$. In (9), $\mathbf{a} \otimes \mathbf{u}$ defines all the possible product combinations of the components of $\mathbf{a}$ and $\mathbf{u}$. The matrix can admit lower- and higher-order polynomial and trigonometric terms [3]. To design the library, first constant and first-order polynomial terms are considered, and then the complexity of the library matrix is increased by introducing higher-order polynomials. Thus, (5) can now be rewritten as

$$\dot{\mathbf{X}}_a = \mathbf{\Upsilon}\mathbf{\Theta}^T(\mathbf{X}_a, \mathbf{U}) \tag{10}$$

which represents the matrix vector product form of (1) where the objective is to discover $\mathbf{g}(.)$ for the ESR in the vast nonlinear function space represented by $\mathbf{\Theta}(\mathbf{X}_a, \mathbf{U})$. In this paper, total variation regularization was applied to obtain the noise-free derivative of $\mathbf{X}_a$ ($\dot{\mathbf{X}}_a$) [23]. For the ESR, $\mathbf{g}(.)$ contains a limited number of terms, and, hence, $\mathbf{\Upsilon}$ is a sparse matrix of dimension $n \times q$.

The least absolute shrinkage and selection operator (LASSO) in [24] works well and promotes sparsity in solving overdetermined system of equations like (10). However, it is computationally expensive for a large dataset. Hence, in this paper, the sequentially threshold least square algorithm was applied to determine $\mathbf{\Upsilon}$ in (10). The algorithm is computationally efficient as compared to LASSO [3].

Once the sparse coefficient matrix $\mathbf{\Upsilon}$ has been determined, the active terms in the ESR can be identified with the indices of the nonzero entries in $\mathbf{\Upsilon}$. In general, the governing DEs of the ESR can be constructed as follows:

$$\dot{\mathbf{a}} = \mathbf{g}(\mathbf{a}, \mathbf{u}) = \mathbf{\Upsilon}^{*T}\mathbf{\Theta}(\mathbf{a}, \mathbf{u}) \tag{11}$$

where $\mathbf{\Theta}(\mathbf{a}, \mathbf{u})$ now defines the vector of the symbolic function of $\mathbf{a}(t)$ and $\mathbf{u(t)}$ whose active terms are governed by the sparse matrix $\mathbf{\Upsilon}^*$. Equation (11) is the matrix vector form of (1), which describes the governing DEs of the ESR.

V. Results

A. Vicon Experiment and Preprocessing

In this experiment, a single sinusoid peristalsis was considered, which is described as

$$H(x,t) = \epsilon + \frac{\alpha}{2}\left[1 - \cos\left(2\pi\frac{x - ct}{w}\right)\right] \quad \text{for,}$$
$$ct \le x \le \left(ct + \frac{w}{2}\right) \tag{12}$$

where ϵ is the minimum radius in mm, x the axial displacement in mm, t the time in s, $H(x,t)$ the conduit's radius at time t, α the amplitude of the wave in mm, c the velocity of the traveling wave in mm/s, and w the wavelength in mm.

The profile describes the nature of the peristalsis that is in the region of $\frac{w}{2}$ because the chambers in front of the wave are in their initial state (not inflated) and behind the wave, they remain inflated, i.e., in their final state. In order to mimic the actual

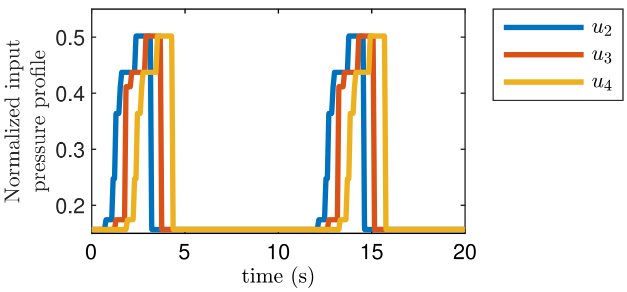


Fig. 4. Normalized input-pressure profile trajectory applied to the ESR's layers (L_2, L_3, and L_4).

biological phenomena of the swallowing process, $\frac{w}{2} = 60$ mm, $c = 30$ mm/s, $\epsilon = 0$ mm, and $\alpha = 10$ mm was considered.

The ESR achieved peristaltic waves of 60 mm half-wavelength by commanding time-varying pressure-profile trajectories into adjacent whorl of chambers simultaneously. These trajectories pass on to a succession along the ESR's length to generate a geometric occlusive propagating wave in the rostral–caudal direction [25]. Fig. 4 illustrates the normalized pressure profile applied to layers L_2, L_3, and L_4 of the robot. In order to profile the ESR's conduit as per (12), the robot was exhaustively searched with different pressure trajectories. It was found that the deformation behavior of the ESR closely resembles that in (12) if it is excited with the control signal shown in Fig. 4. Moreover, one cycle of $\mathbf{u}(t)$ is just a combination of unit-step waveforms of different amplitudes.

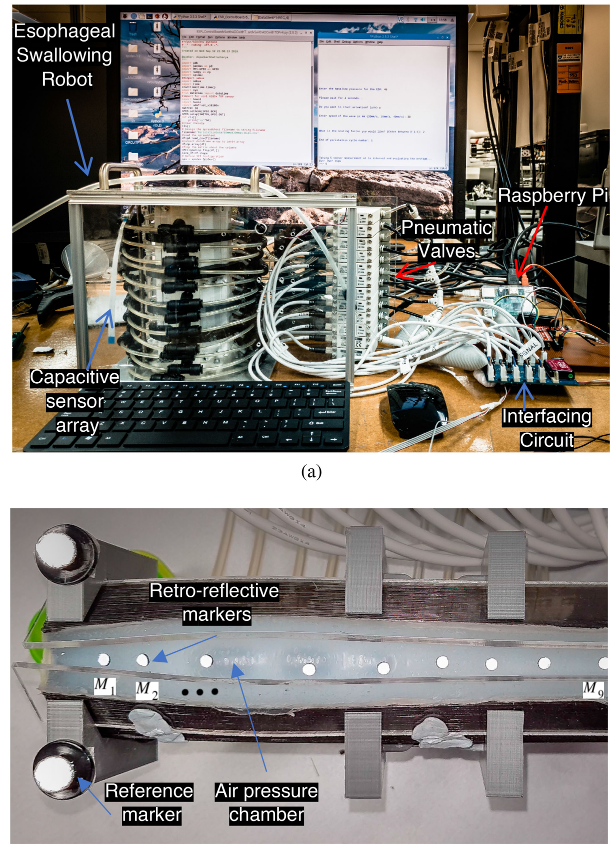
For capturing the surface deformation, 12 Vicon cameras were used to track the motion of the retroreflective markers along the x-, y-, and z-axes, of which four were placed within a range of 2 m from the QSR for precise tracking. From the captured data, it was found that the generated peristalsis profile in the QSR closely resembles the desired profile. Before applying the ROM-based algorithms (see Section V-B), the dataset was preprocessed to interpolate any missing frames in the data, and it was filtered to remove noise from the data. Fig. 5(a) and (b) shows the experimental setup of the ESR and QSR, respectively.

Let $\{M_i\}_{i=1}^{9}$ denote the markers located at the top 9 QSR's layers (see Fig. 3). Let x_{3i-2}, x_{3i-1}, and x_{3i} be the states representing the x-, y-, and z-axes movement of the ith marker, M_i, respectively. Table I provides the description of the states associated with each marker.

By determining the PSD of all the markers states, it was observed that the dynamics governing the robot is identical throughout its length. Hence, instead of taking all the marker states into account for discovering the DEs, the first four markers (M_1, M_2, M_3, and M_4) of the robot were considered. Therefore, from this section onward, the top one-third of the ESR is considered (along its length).

B. Reduced-Order Modeling Results

This section gives the results about mapping the high-dimensional captured snapshot dataset ($\mathbf{X}$) to a low-dimensional subspace S. This section also provides information about the optimal states of the ESR for the DE discovery. The result presented in this section is related to Section III.



(a)

(b)

Fig. 5. (a) ESR with control hardware and valves. (b) Inflated QSR with the arrangement of retroreflective marker.

TABLE I
DESCRIPTION OF THE ROBOT'S STATES ASSOCIATED WITH THE MARKERS

Markers	States		
	x-axis	y-axis	z-axis
M_1	x_1	x_2	x_3
M_2	x_4	x_5	x_6
$\vdots$	$\vdots$		$\vdots$
M_9	x_{25}	x_{26}	x_{27}

To apply the POD, the states of the first four markers were selected, and a snapshot matrix $\mathbf{X}$ was constructed as per (6) such that $\mathbf{X}$ has a dimension of 12×5408 (see Section II). This is illustrated in Fig. 1. The surface plot of $\mathbf{X}$ and its POD approximation $\tilde{\mathbf{X}}$ are shown in Fig. 6. The calculated normalized root mean square error (NRMSE) between $\mathbf{X}$ and $\tilde{\mathbf{X}}$ for $r = 2$ was computed to be 0.1725 mm. Hence, it can be said that $r = 2$ states were enough to represent the maximum variance of the captured dataset.

The marker "o" plot in Fig. 7(a) illustrates the SV energy distribution of the SVD. In Fig. 7(a), the decay of the SVs with the POD modes describes the dominance of each mode. The first three modes, $\{\psi_1, \psi_2, \psi_3\}$, constitute 88% of the total energy in the system. For computational efficiency and convergence of the SINDYc, $\mathbf{\Psi}_r = [\psi_1, \psi_2]$ was chosen. From Fig. 7(b), it can

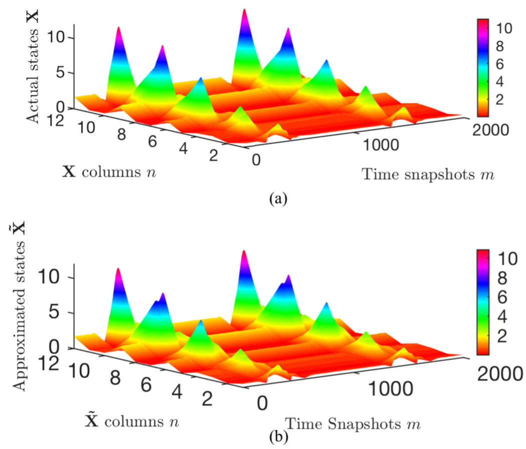


Fig. 6. Comparison between the surface plot of (a) the actual snapshot data matrix $\mathbf{X}$ and (b) its approximated version $\tilde{\mathbf{X}}$.

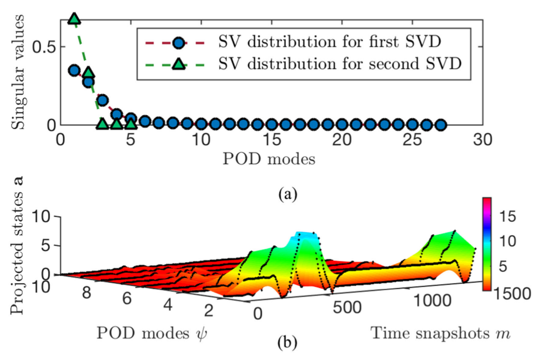


Fig. 7. (a) SV energy distribution associated with the POD modes $\mathbf{\Psi}$. (b) Projection of $\mathbf{X}$ onto the POD modes.

be seen that $\mathbf{X}$ was projected mostly on the first few POD modes to form $\tilde{\mathbf{X}}$. In Fig. 7(b), the magnitude a_j associated with $\boldsymbol{\psi}_j$ is high for $j = 1, 2$ for all time snaps (samples) m as compared to other values of j.

To identify the pivot states of the marker, which are essential for the DE discovery, the OSS algorithm was applied on $\mathbf{\Psi}_r \mathbf{\Psi}_r^T$. The QSR pivoting in the OSS obtained the indices $\varrho = \{12, 6, 9\}$, which implies that $\mathbf{C}$ has unit entries at (1, 12), (2, 6), and (3, 9). By applying $\mathbf{y} = \mathbf{C}\tilde{\mathbf{x}}$, we obtained the pivot states $\tilde{\mathbf{x}}_{12}, \tilde{\mathbf{x}}_6$, and $\tilde{\mathbf{x}}_9$. From Table I, it can be seen that the pivot states $\tilde{\mathbf{x}}_{12}, \tilde{\mathbf{x}}_6$, and $\tilde{\mathbf{x}}_9$ in $\mathbf{y}$ are associated with markers M_4, M_2, and M_3, respectively. Hence, these are the markers that are required for the full-state estimation of the ESR and its DEs discovery [20].

Once the pivot states were determined, a second SVD to $\mathbf{y}$ was applied. In Fig. 7(a), the normalized SVs for both the SVDs were plotted. It was observed that the first two SVs of the second SVD (marker "Δ") plot had captured 95% of the energy, which is a massive improvement in terms of the first SVD. Hence, pivot states also improved the SV response.

C. SINDYc Results

This section presents the DEs of the ESR in the range of S. For applying the SINDYc, the snapshot matrix $\mathbf{X}_a \in \mathbb{R}^{2\times 5408}$ containing the snapshots of $\mathbf{a}(t)$ was considered. By applying $\mathbf{x} \approx \dot{\mathbf{x}} = \mathbf{\Phi}\dot{\mathbf{a}}$, the dataset was projected back to the original states, where $\dot{\mathbf{a}}$ is the prediction (solution) of the discovered DEs. This is illustrated at the bottom of Fig. 1. An input snapshot matrix $\mathbf{U} \in \mathbb{R}^{3\times 5408}$ was also generated for applying the SINDYc such that each row in $\mathbf{U}$ represents the time-varying pressure profile of the L_2, L_3, and L_4 layers of the ESR, respectively (see Fig. 4). A library matrix $\mathbf{\Theta}(\mathbf{X_a}, \mathbf{U})$ containing the candidate terms of the DEs was also generated as per the discussion in Section IV.

The algorithm was controlled by the size of the training dataset, n_{train}, and the sparsity parameter, λ. The values of n_{train} and λ were selected based on the minimum NRMSE criteria and on the number of sparse terms in the DEs. For analysis, eight DE models of the ESR were selected such that each model represented a pair of first-order DEs with various linear and nonlinear polynomial terms (active terms) and different levels of sparsity.

Let $\{H_1, H_2, \cdots, H_8\}$ be the eight DE models of the ESR. Each model is associated with a particular λ and n_{train}. These are the only eight models that are able to validate the test data. The sparse matrix $\mathbf{\Upsilon}^* \in \mathbb{R}^{2\times 252}$ linked with each DE model selects the active terms stored in $\mathbf{\Theta}$ of (11). It was observed that the models did not contain any active term solely formed from the combination of the inputs, and, hence, the DEs did not explicitly depended on the input. The absence of any first- and second-order terms, and the presence of a few third-order terms in the models depicts that the ESR is a highly nonlinear system. In order to select the best model representing the correct DEs for the ESR, further analysis and validation were done on the models (eight models) in Section VI-A. Equations (13) and (14) provide the two discovered DEs for H_5, associated with $n_{\text{train}} = 3400$ and $\lambda = 0.00014$. We have the following:

$$
\begin{aligned}
10^4 \frac{da_1}{dt} &= 2.3a_1{u_3}^2 + 7.6a_2{u_1}^2 + 8.9a_2{u_3}^2 + 40.3a_1^4 \\
&\quad + 140.1a_1^3a_2 - 16a_1^3u_1 + 37.6a_1^2a_2u_1 - 33.5a_1^2a_2u_3 \\
&\quad - 43.5a_1a_2^2u_1 - 6.9a_2^3u_1 - 26a_2^3u_3 - 3.1a_2^2u_1u_3 \\
&\quad - 15.4a_1^3a_2^2 - 38.8a_1^2a_2^3 + 4.8a_1^2a_2^2u_1 - 31.5a_1a_2^4 \\
&\quad - 9.1a_1a_2^3u_1 + 8.6a_1a_2^3u_3 + 43.6a_2^5 + 60.7a_2^4u_3
\end{aligned} \tag{13}
$$

$$
\begin{aligned}
10^4 \frac{da_2}{dt} &= 4a_1^4 + 91.4a_1^3a_2 + 2.7a_1^3u_1 + 66.6a_1^2a_2u_1 \\
&\quad - 68.9a_1^2a_2u_3 - 72.1a_1a_2^2u_1 - 3.6a_1a_2u_1u_2 \\
&\quad + 3.7a_1a_2u_1u_3 + 1.6a_1a_2u_2u_3 + 2.3a_2^3u_1 \\
&\quad - 22.9a_2^3u_3 + 2.8a_2^2u_1u_2 - 3.1a_2^2u_1u_3 \\
&\quad + 1.7a_2^2u_2u_3 - 10.3a_1^4a_2 - 3.3a_1^3a_2u_1 \\
&\quad + 3.6a_1^3a_2u_3 + 2.7a_1^2a_2^2u_1 + 44.2a_1a_2^4 \\
&\quad - 3.9a_1a_2^3u_1 + 3.5a_1a_2^3u_3 + 14.4a_2^5 \\
&\quad + 7.6a_2^4u_1 - 7.2a_2^4u_2.
\end{aligned} \tag{14}
$$

Model H_5 is in the form given by (5). Fig. 8(a) depicts the comparison between the training data $[\hat{\mathbf{a}}^{(\text{train})}]$ and the test data $[\hat{\mathbf{a}}^{(\text{test})}]$ with the predicted data $\dot{\mathbf{a}}$, found by the solution of the DEs defined by H_5. It should be noted that the DEs lie on the

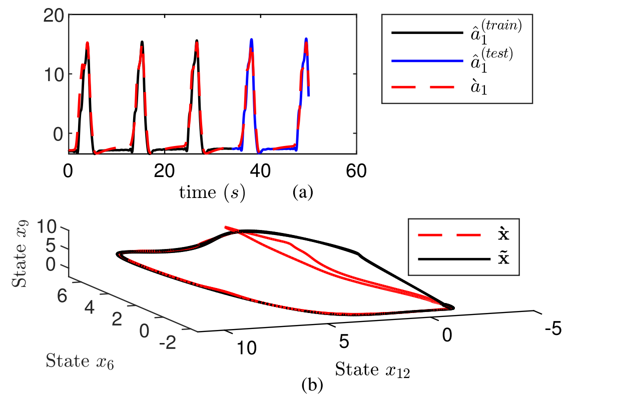


Fig. 8. (a) Comparison between the training data [$\mathbf{a}^{(\mathrm{train})}$] and the test data [$\mathbf{a}^{(\mathrm{test})}$] with the predicted data $\grave{\mathbf{a}}$ determined by the solution of the DEs of H_5. (b) Comparison between the state trajectories obtained from the actual states and the solution of the DEs.

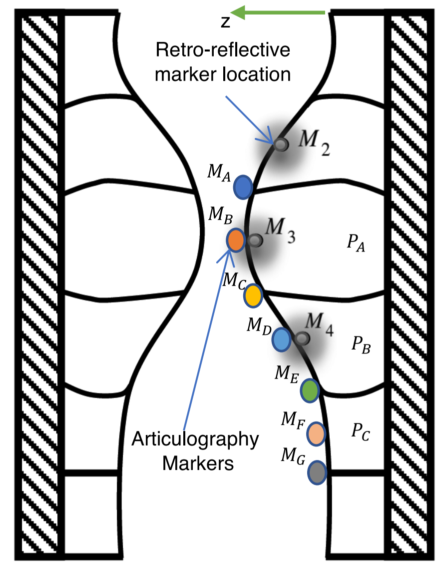


Fig. 9. Schematic of the articulography experiment setup [12].

range of $\mathbf{\Psi_r}$ and not on the standard basis space. Once the DEs were determined, their predicted values were projected back to the S_p by applying the transformation matrix $\mathbf{\Phi}$ defined in Section III-C. Fig. 8(b) illustrates the comparison between the deformation trajectories defined by the actual (captured deformation) and predicted states (predicted deformation). In general, by premultiplying $\mathbf{\Phi}$ with the solutions of (13) and (14), the deformation of the ESR with respect to a specific input $\mathbf{u}$ can be predicted.

VI. Validation of the Robot's DEs

This section validates the eight DE models of the ESR, determined in Section V-C with the articulography experiments in [12] for the purpose of selection of the best model. The stretchable array of the capacitive sensor results in [26] was also utilized to further validate the selected DE model.

A. Articulography Tests

Fig. 9 illustrates the articulography setup on the ESR. P_A, P_B, and P_C are the pneumatic pressure values exerted in triads over the range of 0–71.5 kPa, divided into 20 steps. In Fig. 10(c), u_1, u_2, and u_3 represent the profiles of P_A, P_B, and P_C such that $u_1 \geq u_2 \geq u_3$. Each commanded pressure triad was held for 1 s so that the system could reach its steady-state condition, and, hence, the time step used for solving the eight models of the DEs was also taken to be 1 s. The input-pressure-trajectory, as shown in Fig. 10(c), consists of the original trajectory captured between $t = 300$ and 600 s.

To simulate the DE models, a random set of initial conditions was used. From the articulography experiment, the marker displacements M_A, M_B, and M_D were selected to validate the results of the DEs simulation. The given set of markers was selected because it was placed close to the location at which the Vicon markers were located.

After simulating all the models with the input-pressure profile used for performing the articulography experiments [see Fig. 10(d)], it was found that H_5 best predicted the output. When subjected to the input described in Fig. 10(d), the blue and red color plots in Fig. 10(a), (b), and (c) show the responses of the articulography markers and the discovered DEs, respectively. Both the responses are very similar to each other, especially at the peaks indicated by the error bar plots, as shown in Fig. 10(a), (b), and (c).

Let $\mathbf{E} = [\mathbf{e}_1(t), \mathbf{e}_2(t), \mathbf{e}_3(t)] \in \mathbb{R}^{301\times 3}$ be an error matrix, such that $\mathbf{e}_1(t) = (M_A - \grave{x}_6), \mathbf{e}_2(t) = (M_B - \grave{x}_9)$, and $\mathbf{e}_3(t) = (M_D - \grave{x}_{12})$. Each row of $\mathbf{E}$ represents the error between the articulography marker (blue color) and DEs predicted displacement (red color) from $t = 300$ to 600 s. The mean and standard deviations of $\mathbf{E}$ were found to be $[-0.0584, 0.9852, 0.9397]$ and [0.2047, 0.2658, 0.2633], respectively. Since the standard deviations of the columns of $\mathbf{E}$ are low, no significant change in $\mathbf{e}_1(t), \mathbf{e}_2(t)$, and $\mathbf{e}_3(t)$ with time was detected. In addition, from Fig. 10, it can be seen that the distance between the peaks (height of the error bars) are approximately equal at the indicated time instants. Overall, $\mathbf{E}$ does not affect the prediction of the DEs because $\mathbf{e}_1(t), \mathbf{e}_2(t)$, and $\mathbf{e}_3(t)$ can be considered as offsets.

The errors $\mathbf{e}_1(t), \mathbf{e}_2(t)$, and $\mathbf{e}_3(t)$ (error at the peaks) at $t = 366$ s, $t = 457$ s, and $t = 564$ s were found to be $[-0.04, 0.84, 0.51]$, $[-0.19, 0.79, 1.15]$, and $[-0.17, 0.85, 1.59]$, respectively. In Fig. 10(a)–(c), errors $\mathbf{e}_1(t), \mathbf{e}_2(t)$, and $\mathbf{e}_3(t)$ occur due to the slight mismatch in the location of the articulograph and the Vicon markers, which has been shown in Fig. 9. Moreover, since the data from both the experiments were recorded by using different measurement systems, so they introduce different kinds of measurement errors. Interestingly, the DEs predicted plot retains most of the characteristics of the articulography plot.

Fig. 10(a)–(c) reveals that the sawtooth variations recorded by the articulograph are comparatively fewer than the variations generated by the H_5 DEs prediction. This is because of the inability of the articulograph to capture the variations accurately, and because of the effect of the noise while recording. Interestingly, it can be observed from Fig. 10(a)–(c) that the DEs' response jumped toward the response recorded by the

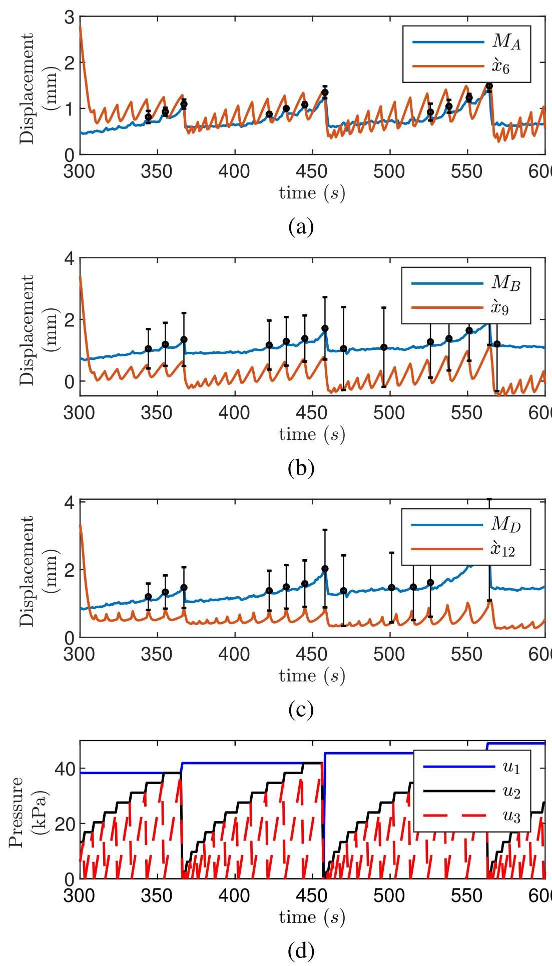


Fig. 10. (a), (b), and (c) Plots comparing the articulography markers displacement, and the discovered DEs' prediction, when subjected to the input-pressure profile in (d).

articulograph, irrespective of the fact that random initial conditions were selected for solving the DEs.

B. Capacitive Sensor Array Tests

A stretchable array of capacitive sensors was implanted in the ESR's conduit [26]. The array was used to measure the conduit's deformation at layers L_6, L_7, and L_8 (see Fig. 2). The result from the array experiments was used to further validate the DEs prediction because they were discovered from the data collected from the QSR.

The pressure-trajectory profile responsible for generating the swallowing action (peristalsis waves) in both the sensor array and the Vicon experiments was kept the same. Since the ESR was configured to generate uniform peristalsis characteristic across the conduit, so it was assumed that the action produces similar deformation profile at any point on the conduit wall. For this reason, the deformation states of layers L_6, L_7, and L_8 of the sensor array experiments were compared to the DEs' prediction associated with layers L_2, L_3, and L_4.

Fig. 11 illustrates the deformation trajectory obtained from the sensor array experiments (dashed line), and from the simulation of the DEs (solid line). The deformation obtained from DEs' prediction was higher as compared to the sensor array results. This is because a manometry catheter of diameter 4 mm was inserted in the conduit of the ESR while performing the sensor array experiments, due to which the conduit's chambers were not able to attain their maximum deformation. Hence, a maximum deformation of 8 mm was recorded by the sensor array.

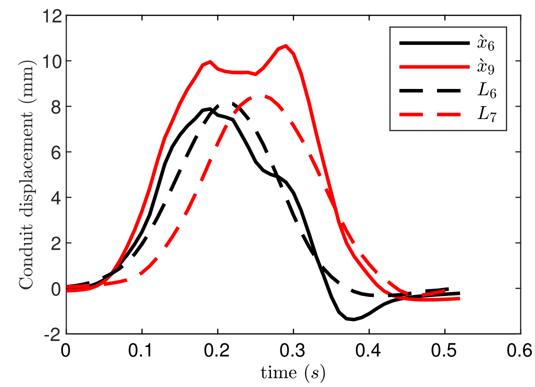


Fig. 11. Comparison between the state trajectories obtained from the capacitive sensor array experiments and the simulation of the DEs.

The maximum deformation difference further goes high because for any layer in the ESR, the chamber deformation measured at the center is restricted from the inflation of another chamber in front of it. In the QSR, due to no such restriction, the deformation of some of the layers went past the 10 mm mark.

The mismatch between the profiles of the trajectories might be attributed to the limitation of the sensor array in capturing the dynamics of the ESR precisely because the sensor was designed, mainly, to record the maximum deformation. The high accuracy in the DEs' prediction is also evident of the fact that the Vicon system is a more capable system in terms of sensing the ESR's conduit deformation.

VII. Conclusion

This paper discussed the discovery of DEs for a soft-bodied ESR. The ML techniques applied to determine the DEs rely solely on the data collected from the experiments. The DEs modeled the ESR's nonlinear peristaltic conduit deformation with respect to the applied pressure. The discovery of the DEs has given significant rise to the understanding of the complex and nonlinear dynamical behavior of the ESR.

To collect experimental dataset, a series of swallowing experiments was performed on the robot. The dataset consisted of time-varying conduit deformation, which were measured using the Vicon system. The POD was applied to reduce the dimension of the dataset. The reduced dataset was, then, used to determine the DEs of the ESR by applying the SINDYc. The presented strategy makes several noteworthy contributions to the ESR and, hence, to the area of soft robotics.

In the future, the DEs will be used to design the closed-loop control system for ESRs. The addition of the controller will extend the biomimetic capability of the ESR, which will make it compliant with different food boluses. Hence, more complex swallowing behavior comprising food bolus condition will be simulated.

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